

Experiment 17: Sum of Integers from 1 to n

1. Aim

To write and execute a Prolog program to find the sum of integers from **1 to N** using recursion.

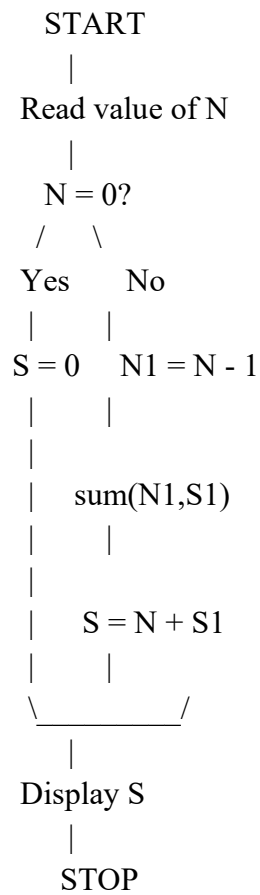
2. Objective

- To understand recursion in Prolog.
- To calculate the sum of integers from 1 to a given number N.
- To learn how to define facts and rules in Prolog.
- To execute queries and obtain the required result.

3. Algorithm

1. Start.
2. Define the base case: when $N = 0$, the sum is 0.
3. For $N > 0$, calculate $N - 1$.
4. Recursively find the sum from 1 to $N - 1$.
5. Add N to the previous sum.
6. Store the result in S.
7. Display the sum.
8. Stop.

4. Flowchart



5. Prolog Program

sum(0, 0).

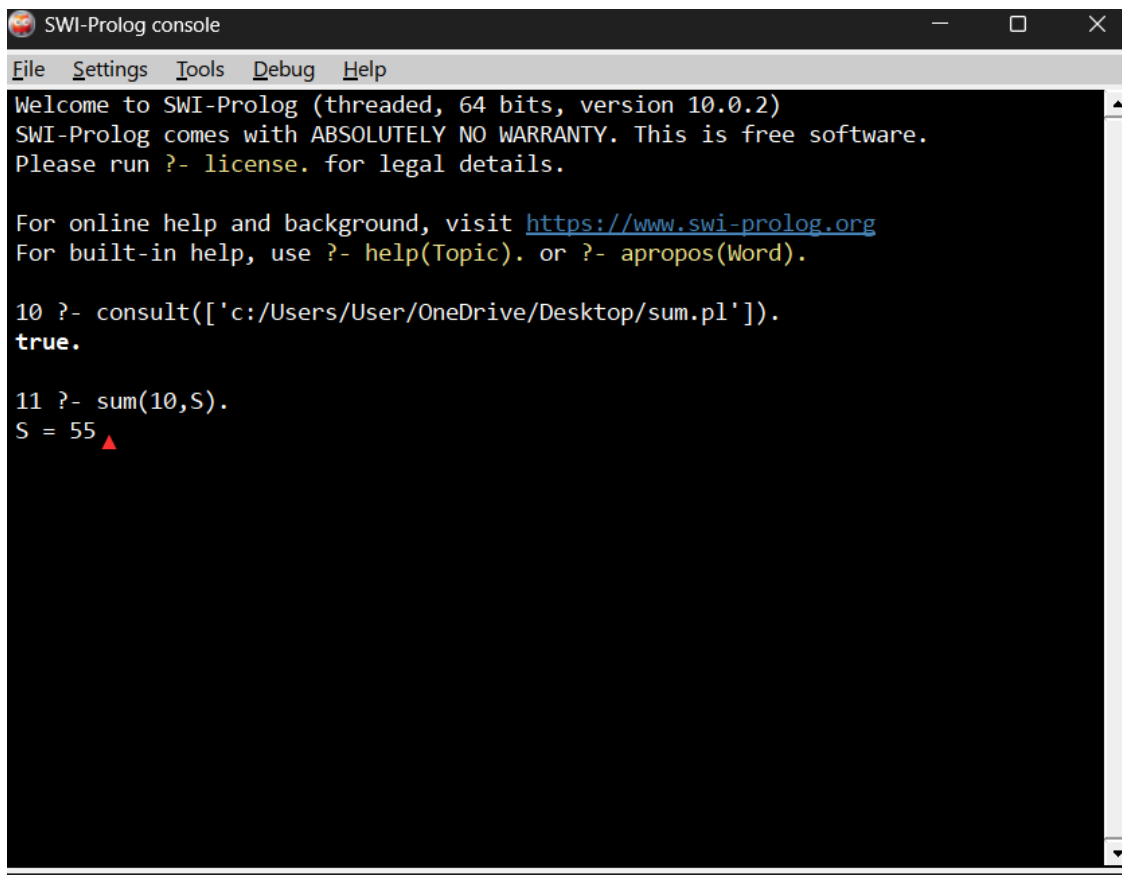
sum(N, S) :-
 N > 0,
 N1 is N - 1,
 sum(N1, S1),
 S is N + S1.

Query

To find the sum of integers from **1 to 10**, enter:

```
?- sum(10,S).
```

6. Sample Output

A screenshot of the SWI-Prolog console window. The window has a title bar "SWI-Prolog console" and a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The console text is as follows:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

10 ?- consult(['c:/Users/User/OneDrive/Desktop/sum.pl']).
true.

11 ?- sum(10,S).
S = 55 ▲
```

7. Result

Thus, the Prolog program to calculate the **sum of integers from 1 to N** was successfully written and executed using recursion. The sum for the given value of N was obtained correctly.

8. Conclusion

The experiment demonstrates the use of **recursion in Prolog** to solve a mathematical problem. The program repeatedly reduces N by 1 and adds each value until it reaches the base case 0.